

How to prepare knowledge for your AI Agent?

Background

Recently many user asked me about how to prepare knowledge for their agents? how to write the knowledge documents for agent? What characteristics should good knowledge have? let's find a solution for these question in this articles.

Why we need upload knowledge to AI Agent?

Traditional LLMs have three major pain points, which Smart provides solutions through knowledge base integration:

- 1. **Insufficient Knowledge Timeliness:** LLMs' knowledge is limited by the deadlines of training data (for example, GPT-3.5 is only updated until 2021), making them unable to answer real-time questions such as new policies and technological advances. RAG ensures that generated content remains current by connecting to dynamically updated external knowledge bases (such as the latest research reports and corporate documents).
- 2. **Hallucination Problem:** When LLM encounters knowledge gaps, it may generate seemingly plausible but in fact inaccurate content. RAG significantly reduces the risk of hallucination by incorporating authoritative external information (such as Wikipedia and professional databases) to force the model to generate answers based on real data.
- 3. **Lack of Access to Private or Specialized Domain Knowledge:** LLM struggles to cover internal corporate data (such as internal system user guide) or highly specialized knowledge (such as Shopee Help Center Articles). RAG allows for connection to private knowledge bases, providing domain-accurate answers without uploading sensitive data or retraining the model.

What Kind of Knowledge I can use?

In our daily work, we generated a lot of documents, such as:

Document Type	Sub Type	Documentation Tools				
		Google Doc	Google Slides	Google Sheet	Confluence	Other System
Product Documents	BRD / FRF	Used	Used	Used	Used	/
	PRD	Used	/	Used	Used	/
	Design	/	/	/	/	Figma
	User Guide	Used	Used	/	Used	Data Suite Forum
	Training Sessions	/	Used	/	/	/
	Public Articles	/	/	/	/	Help Center Articles: CS System Seller Education Hub: Seller-CMS Admin
Technical Documents	Technical Design	Used	/	/	Used	/
	DoD SOP	/	/	/	Used	/
project Documents	Request Management	/	/	Used	/	POP, Jira
	Meeting Summary	Used	/	/	Used	/

Usually, we **recommended user to upload existing documents to save their efforts**. But we found that some documents did not perform as expected.

How to prepare knowledge?

First of all, let me introduce the key processing steps after user uploaded their documents.

1. Parsing: Convert different documents into standardized text, extract original text, image, tables from uploaded documents.
2. Chunking: Cut documents into semantically complete chunks to adapt to model input constraints and improve retrieval accuracy.
3. Retrieve: Recall the most relevant snippets to the query from the knowledge base to provide accurate knowledge for answer generation.

Parsing

Currently Smart support users to upload 12 types of documents: docx, txt, csv, xls, xlsx, pptx, PDF, MD, JPEG, JPG, PNG, SVG. Smart also support user upload their knowledge from Google Drive, Confluences, Hive Table(through Data Service API), and provide Website Crawler to extract content from websites(no authentication required website only)

- Documents from Google Drive: Smart will convert Google Doc to docx type, Google Sheet to xlsx type, Google Slides to pptx type.
- Documents from Confluence: Smart will convert confluence page to Markdown.

We classify these documents into four categories:

	Categories	Document Types
1	Document	docx, txt, PDF, MD
2	Table	csv, xls, xlsx
3	Slides	pptx
4	Image	JPEG, JPG, PNG, SVG

For Document Category, the completeness of the content after parsing is **txt > MD > docx > PDF**.

- txt, MD and docx are all structured text, the text content will be fully extracted, style information (such as comments, hyperlink locations) may be lost or reduced to plain text
- Because PDF stores image information rather than structured text, it requires OCR, layout recognizer and table recognizer to reconstruct content and formatting. Information may be incorrect or lost during the parsing process.
- When you export the document and want upload it to Smart, please try to choose to export it in txt/Markdown/docx format instead of PDF, so that Smart can get more accurate content.
- If you want to experience the PDF parsing results, you can test it here: <https://compass.llm.shopee.io/playground/extract>

For Table Category, since csv, xls, xlsx are all structured text, the text content will be fully extracted too

For Image Category, Smart will use LLM model to understand the image and generate descriptions for each images by default. You can select a better LLM Model when uploading to get better understanding and description. and also you can toggle ON OCR function when uploading, Smart will run another OCR service to extract text from images.

Parsing Summary, **Markdown / Docx / txt** can be extracted **more complete** content in document parsing **than PDF / Slides.**

We have previously introduced why chunking is needed. now we are talking about how to chunk.

There are two common chunking methods: one is to **split according to the header**, another is to **split according to the delimiter**.

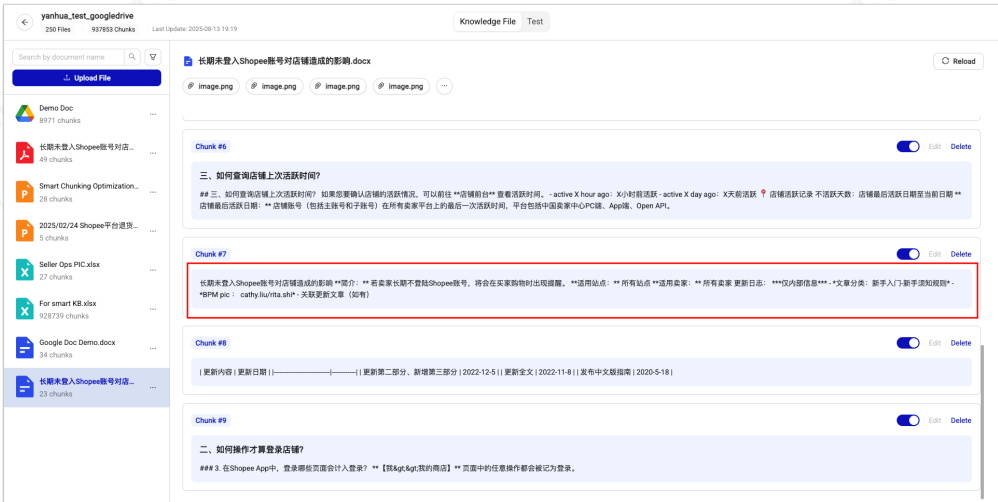
Most of the documents produced during work have a clear structure and format, we suggest users to select default chunking strategy or split by header strategy. (actually default chunking strategy is split by header but without custom parameters)

But we still got some issue feedback on chunking, We also provided corresponding suggestions for these issues.

Issue 1: Chunks are cut too small, resulting in incomplete context

If it's splited by header, Try to select one higher level header to chunking, Smart won't split the header which level lower than selected level, so the content will be more complete

Issue 2: My document has a separate paragraph describing the background and scope of application, but this part was cut into a separate chunk, resulting in the lack of background and scope of application in my content.



Currently regardless of the slicing method used, the content in different paragraphs of the document will be cut into different chunks. If you want these contents to remain in the same fragment, please try to write them under the same paragraph or heading.

Meantime, Smart is exploring a Meta Data feature that supports users to set meta data and tags for chunks, which can provide more information for LLM beyond the document.

Issue 3: There are some high-quality articles in the internal system and I want to import them into KB. How can I do that?

Currently, Smart supports importing content from other website systems in two ways, Website Crawler and Hive Table.

If this website does not require authentication, you can use Website Crawler and toggle ON the Scrape All, Smart will recursively visit each link inside the page and extract content and sub pages from it.

However, For some client side rendering websites or websites with complex structure, it's hard for website crawler to get the full page structure and content, sometimes it will miss some sub pages. that is why we suggest user that it is better to import data into Smart by database synchronization rather than Website Crawler.

If you are the system owner, you can sync your articles table to Hive and create a Data Service API based on the table, then you can upload the table content to Smart by Data Service API.

If you are not the system owner and want to use it's knowledge, please contact Smart team to find a solution to upload it.

Retrieve

After all the knowledge has been uploaded and added to the Agent, the Agent will recall relevant chunks from the knowledge base based on the user input and then generate a response.

But sometimes the retrieved results are not as expected, which may be due to the following reasons:

1. There are many similar knowledge chunks in the knowledge base, and they have slight differences in some aspects.
2. The language of the knowledge base is different from the language of the user's questions
3. The user's question contains some abbreviations, which makes it hard to match the content correctly

If your knowledge base contains a large amount of similar content, we recommend splitting it into different knowledge bases based on their conditions, and then prompting LLM to call different knowledge bases in different situations to obtain more accurate results.

If the user's question contains abbreviation or language problems, you can prompt LLM with the full meaning of some abbreviations in the prompt and ask LLM to expand and translate them into the knowledge base language before searching.